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AMD Ryzen 9 3900X

Zen 2 knocks it out of the park

The 3rd Gen Ryzen processors have brought about a slew of improvements across the SKU stack. The top SKU at launch, i.e. the Ryzen 9 3900X has 12 cores and a massive L3 cache, uses the same AM4 socket, is compatible with previous gen platforms. The cost, however, has equalled Intel SKUs as both currently sell for about Rs.42K. If we are to compare launch prices, then the Intel Core i9-9900K was selling for Rs.64K in India with barely any stocks available anywhere. With the Zen2 launch, there were a couple of hiccups along the way. We had to retest the processors because of issues with the initial BIOS that came with the boards weren't optimal. Let's see how these new CPUs fare.



PERFORMANCE.....90
VALUE FOR MONEY.....75

but it also bumped up the TDP of the PCH module. So all new X570 motherboards with the exception of a few have active cooling for the PCH. Speaking of motherboards, if you are running the Ryzen 7 or Ryzen 9 processors, you

will have to be careful about which motherboard you pick since not many have VRM circuitry that can handle the extra power load that the top Ryzen 3rd Gen processors tend to consume. Overclocking is out of the question with a lot of the motherboards. While the complexities of picking the right motherboard would require another article altogether, it can be simply put forth that if you're going to be running a 3700X or 3800X on stock, then your VRM should be able to provide 75A on Vcore without issue. For the 3900X and the upcoming 3950X, you should aim at 100A Vcore. If you're OCing these processors, just double the Vcore amperage that would be required from the motherboard.

Performance

As usual, we begin our benchmarks with Cinebench R15 and the Ryzen 9 3900X comes short of the Intel Core i9-9900K by a small fraction in single-threaded runs but the multi-threaded runs is ahead of the 9900K by about

50% which is a massive improvement. Compared to the older 2700X, the 3900X has made a significant improvement in single threaded performance. This would be a Zen+ to Zen2 comparison. When you bring the 1800X into the picture then the performance gap widens even more.

In our Handbrake encoding tests, the Ryzen 9 3900X beats the 9900K across all tests. And the results remain the same with our 7-zip and WinRAR compression/decompression tests as well.

Then comes gaming, a domain that Intel has had the lead for quite some time. In our gaming benchmarks, except for Strange Brigade and Shadow of the Tomb Raider, the 9900K was leading in all games. We even saw the 9700K perform better than the 3900X in quite a few benchmarks. So gaming is where AMD still has some catching up to do.

We managed to OC the processor to about 4.2 GHz at 1.35V which is a decent upgrade over stock but we couldn't run all the relevant benchmarks given in the limited time we had with the processor. Having to retest after the BIOS updates took up a considerable amount of time.

Verdict

The improvement in single-threaded, multi-threaded and gaming performance have all been noteworthy. The only issue here lies with picking the right motherboard. It should be noted that for folks opting for the lower-end processors, this won't be an issue. Overall, the Ryzen 9 3900X has tilted the scales in AMD's favour and we can't wait to see how the 10th Gen Intel Comet Lake-S processors will perform.

—Mithun Mohandas

Features

The Ryzen 9 3900X makes use of a chiplet architecture with 2 CPU chiplets and a third one for the I/O. This design gives it 64 MB of L3 cache whereas all the other processors in the 3rd Gen family have 32 MB of L3 cache. The upcoming 3950X will also have 64 MB of cache. Compared to the previous desktop top SKU (Ryzen 7 2700X), the Ryzen 9 3900X has gained 50% more cores, however, the TDP remains the same. Most of this savings are attributed to the improved TSMC 7nm process which Zen2 uses. Even the clock speeds are higher, if only by 100 MHz. The usual norm is to see a drop in clock speeds with the addition of more cores since the power requirements go through the roof and pushing higher clocks becomes difficult. With Zen2, we see improvements on both accounts.

The other improvements involve the inclusion of Gen 4 PCIe. This allows motherboard manufacturers to have more peripherals connected

SPECIFICATIONS

CORES: 12 | THREADS: 24 | BASE CLOCK: 3.8 GHz | BOOST CLOCK: 4.6 GHz | PROCESS NODE: TSMC 7nm | L3 CACHE: 64 MB | L2 CACHE: 6 MB | TDP: 105 W.

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